

From Randomness to Pattern: Exploring the Plinko Distribution

PhET Plinko Probability · PhET

Years 9-10	~30 minutes	Mathematics · Statistics & Probability
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Simulation: <https://phet.colorado.edu/en/simulations/plinko-probability>

Curriculum links: AC9M9ST04 · AC9M10P02 · AC9M10ST02

Learning intentions

- I can run repeated trials in a simulation and record the resulting distribution.
- I can compare an experimental distribution to the theoretical binomial distribution.
- I can describe a distribution's shape, centre and spread in context, and justify a display choice.

Getting started

Open Plinko Probability and start by dropping a small number of balls (e.g. 10) through the pegs, watching where they land in the bins. Then switch to running a much larger number of trials and turn on the histogram and theoretical-distribution overlay for the later tasks.

Tasks

1. Drop 10 balls and describe the shape of the pattern that forms across the bins.

2. Run the simulation at three increasing ball counts (e.g. 10, 100, 1000) and record the overall shape of the histogram each time.

Number of balls dropped	Shape of the resulting histogram

3. Based on your table, describe what happens to the pattern as the number of balls increases.

4. Turn on the theoretical distribution overlay and compare it with your experimental histogram at the largest ball count. Describe how closely they match and explain why more trials help.

5. Change the number of rows of pegs. Describe how the shape and spread of the resulting distribution changes.

6. Explain why this simulation is described as modelling a 'binomial' situation, referring to what happens to a ball at each individual peg.

7. A histogram is one way to display this data. Justify why a histogram is an appropriate display choice for Plinko results, compared with, say, a simple list of bin counts.

Extension (optional)

This Plinko model behaves similarly to repeated coin flips. Describe another real-world repeated-chance scenario that would produce a similar bell-shaped distribution, and explain why.
